Rank-Aware Hyperbolic Alignment for Vision-Language Dataset Distillation

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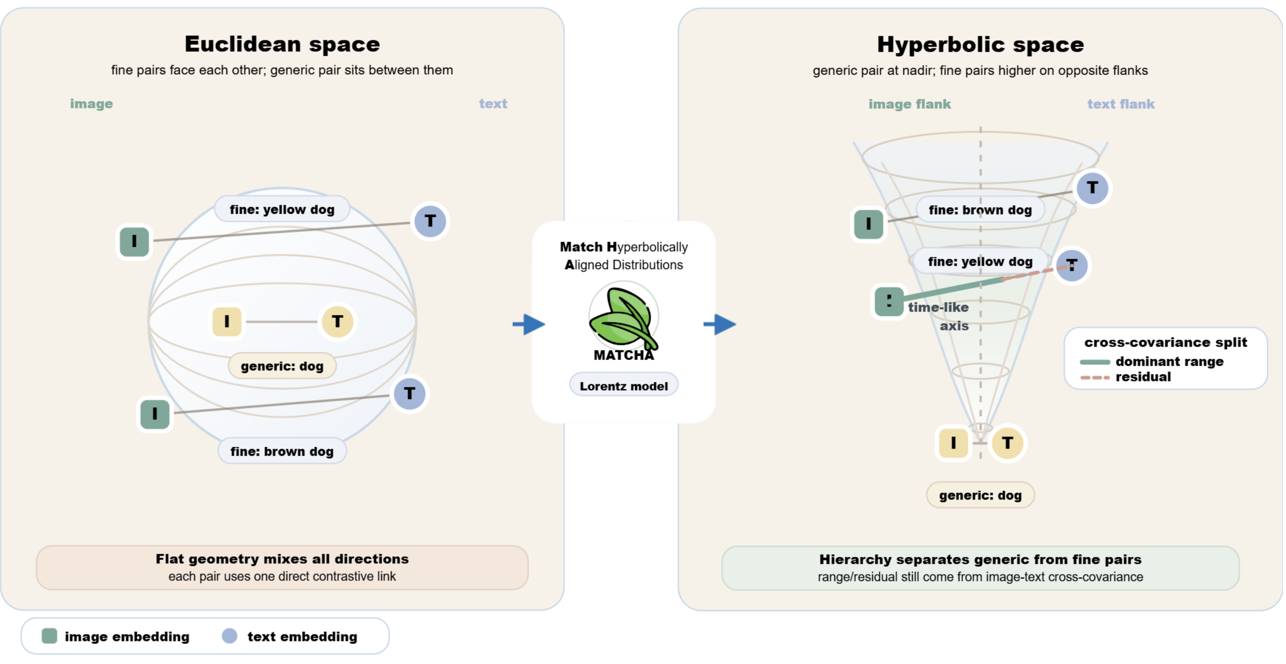
Introduction

- Demand for multimodal datasets for efficient training of VLMs is growing!
- Existing vision-language dataset distillations preserve *cross-correlation* structure, yet miss explicit pairwise structure and does not consider the hierarchy in text

→ RAHA: relative similarity order + hierarchical structure

Motivation

Transition to Hyperbolic Space:



What must data compression preserve?

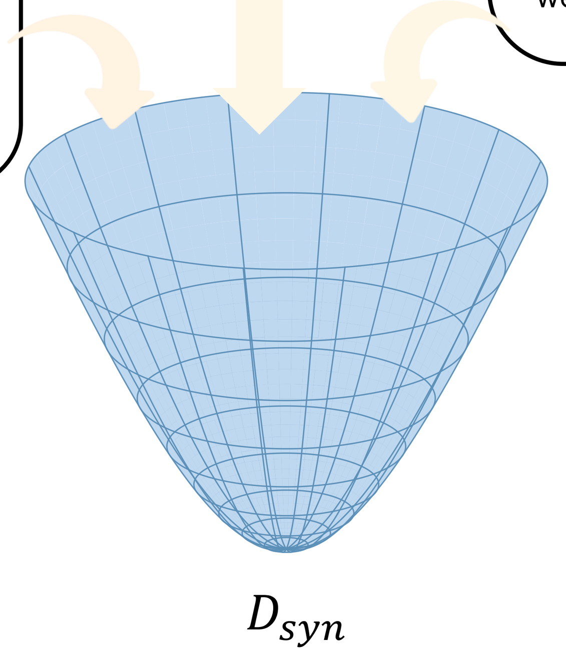
- A distilled image-text set is useful only if a model trained on it learns the same bidirectional ranking behavior as from real data. This makes cross-modal relevance the core information capacity to preserve, beyond visual appearance alone.

How should text hierarchy be retained?

- Since captions move from generic categories to fine attributes and relations, a geometry that keeps generic text broad, and fine-grained distinct on a hyperboloid is beneficial

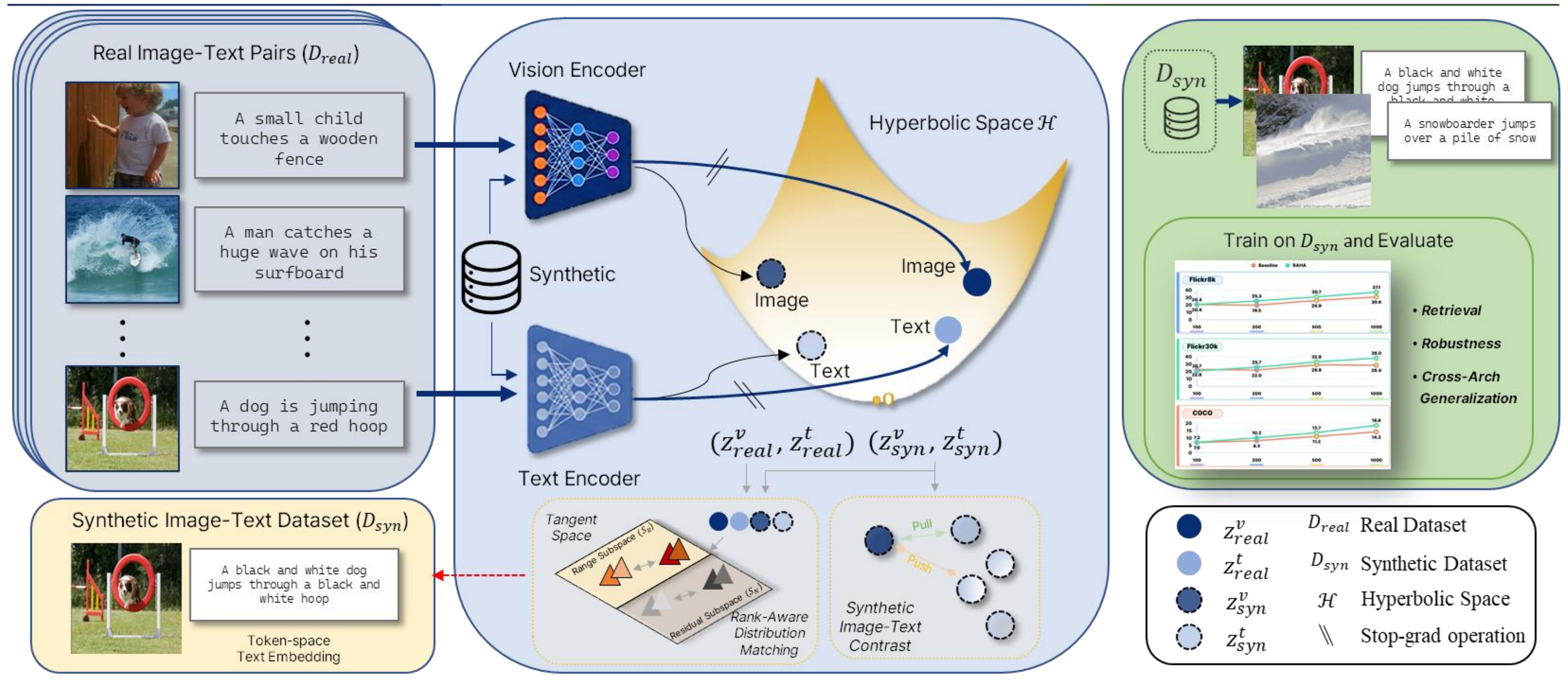
Where should data capacity be spent?

- Previous study has shown that useful image-text relations are concentrated in low-rank structure. In RAHA, we separates range and residual subspaces, matching dominant shared semantics while regulating weaker residual variation.



Proposed Method: RAHA

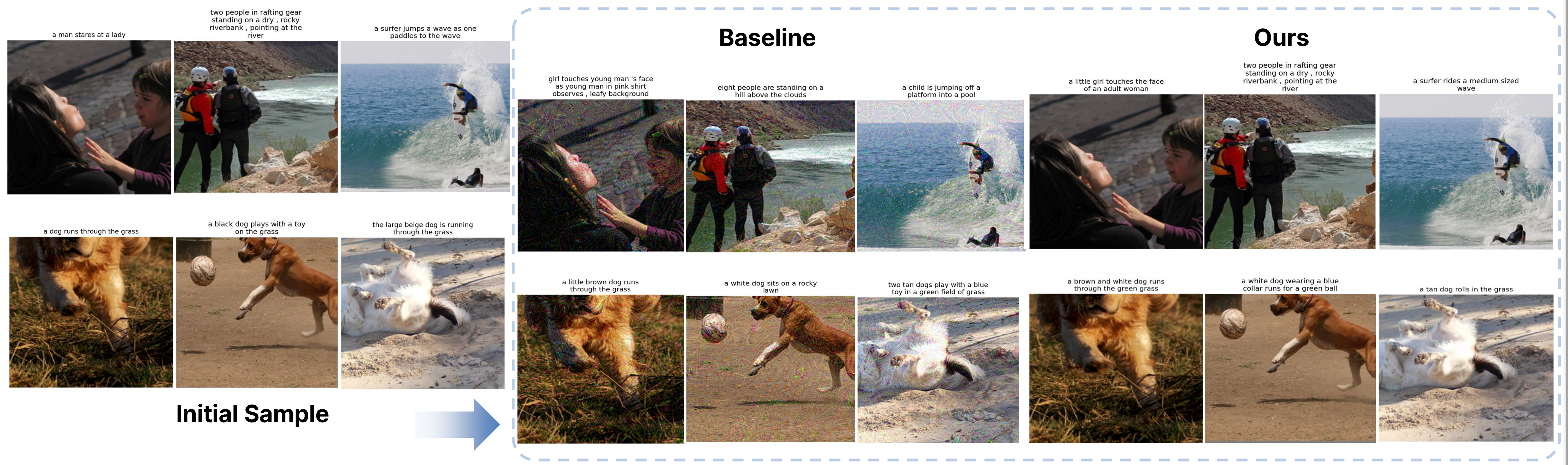
Match Hyperbolically Aligned Distributions!



Key Contributions

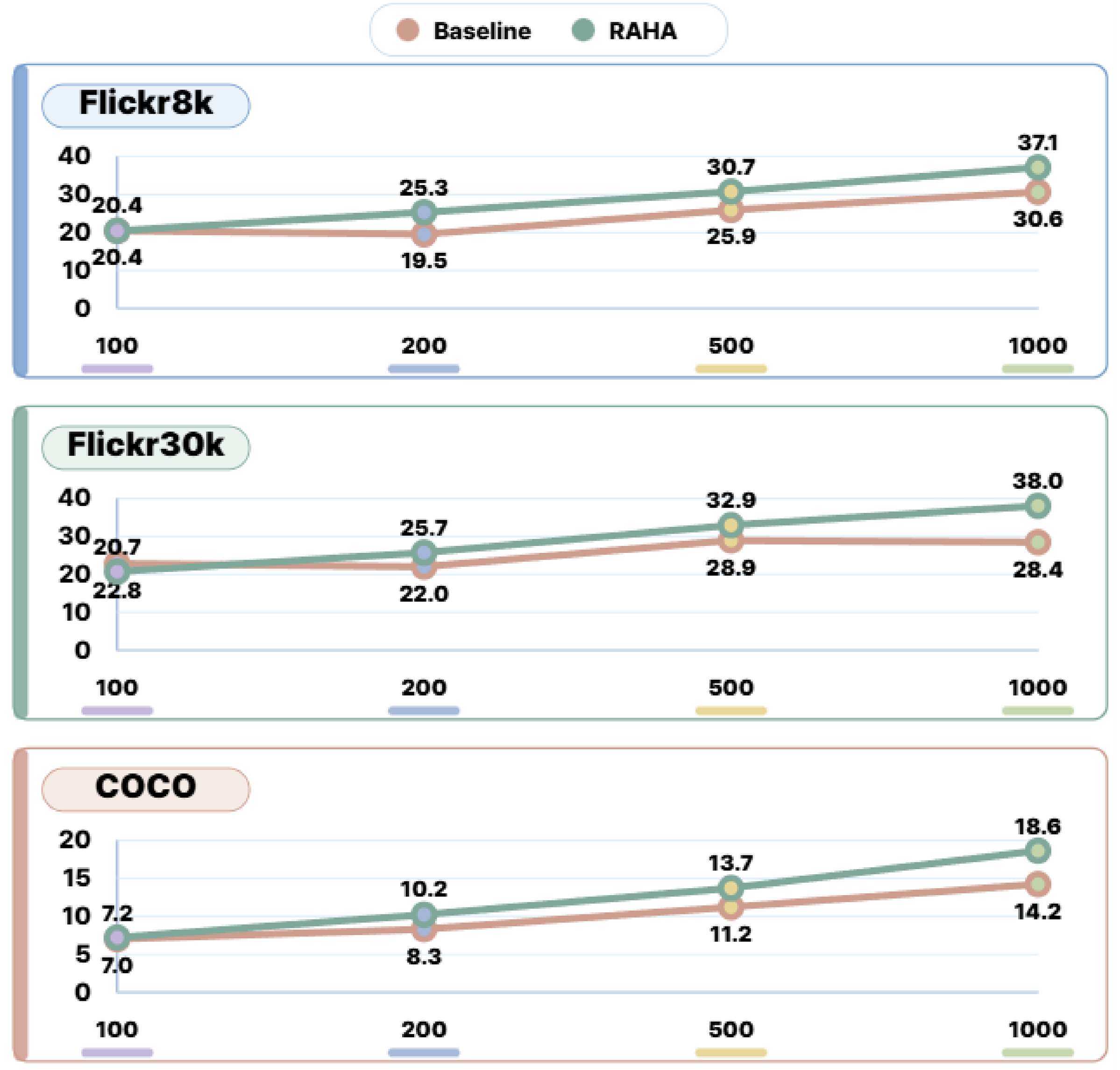
1. Rank-aware hyperbolic formulation for vision-language dataset distillation
2. Range-residual subspace-decomposition for cross-modal relevance distillation with optimal transport
3. Comprehensive evaluation on retrieval, cross-arch. transfer, robustness, prompted classification

Qualitative Comparison



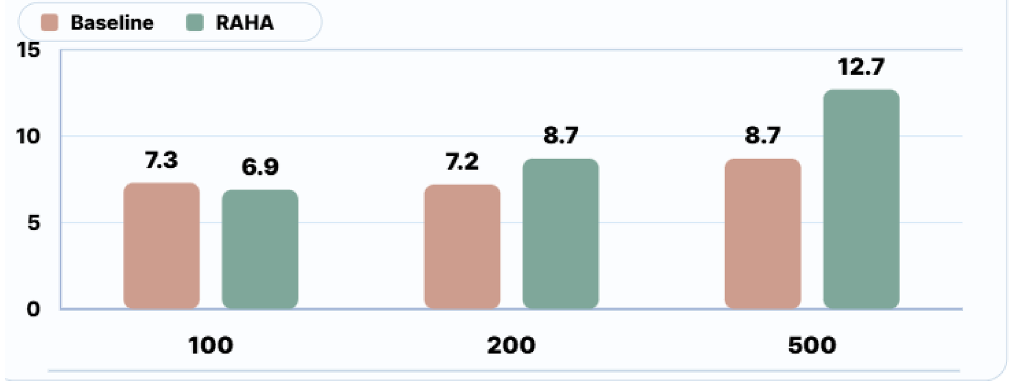
Experiments

Image-Text Retrieval



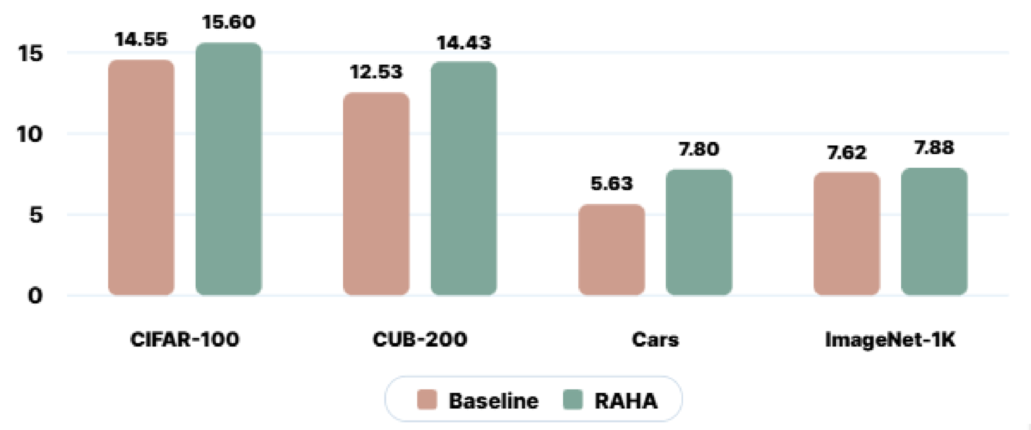
DATASET	PAIRS	BASELINE MEAN	RAHA MEAN	Δ
Flickr8k	100	20.4	20.4	+0.0
	200	19.5	25.3	+5.8
	500	25.9	30.7	+4.8
	1000	30.6	37.1	+6.5
Flickr30k	100	22.8	20.7	-2.1
	200	22.0	25.7	+3.7
	500	28.9	32.9	+4.0
	1000	28.4	38.0	+9.6
COCO	100	7.0	7.2	+0.2
	200	8.3	10.2	+1.9
	500	11.2	13.7	+2.5
	1000	14.2	18.6	+4.4

Cross-Architecture Transfer

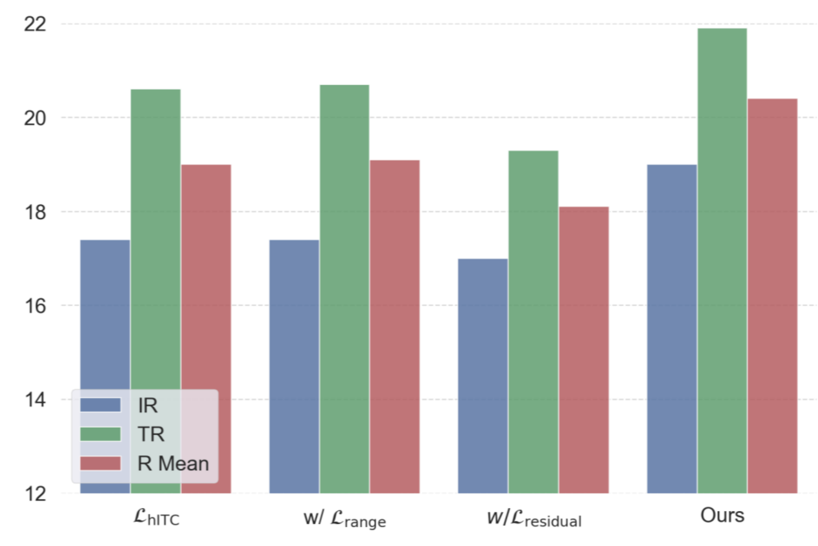


*Averaged over {NF-ResNet, NF-RegNet, ViT-B}, {BERT, DistilBERT} w/o source model

Prompted Image Classification



Component Analysis



Robustness to Noise

(a) JPEG (75%)/Bit Quantization (4-bit)
(b) Additive White Gaussian Noise (AWGN) (σ = 0.01)
(c) 10-PGD (step size=2)

		Image-side δ				Text-side δ			
Pairs	Method	(a)	(b)	(c)	Mean	(a)	(b)	(c)	Mean
100	Baseline	6.0	6.0	1.9	4.6	3.8	10.1	0.0	4.6
	Ours	4.9	5.6	1.8	4.1	3.1	8.1	0.0	3.7
200	Baseline	5.0	5.7	1.9	4.2	3.3	8.7	0.0	4.0
	Ours	5.8	6.8	2.4	5.0	3.6	9.8	0.0	4.5
500	Baseline	8.0	8.3	3.0	6.4	3.9	11.5	0.0	5.1
	Ours	8.0	9.5	3.0	6.8	4.4	15.0	0.0	6.4